

Database Schema Notes

This document summarizes the main database structure used in the shift management system.

The actual production schema contains additional constraints and operational fields, but this document focuses on the core design relevant to the system architecture.

Core Store Structure

stores

Represents each store using the system.

Main responsibilities:

- Store identity
- Store-specific URL slug
- Staff position mode
- Manager / staff access separation

Related tables:

- `store_staff`
- `periods`
- `store_business_hours`
- `store_required_staff_settings`
- `auto_shift_runs`

store_staff

Represents staff members belonging to each store.

Main responsibilities:

- Staff display name
- Sort order
- New staff flag
- Manager flag
- Position type: `common`, `kitchen`, or `hall`

This table is used by both normal shift management and Auto Shift V1.

periods

Represents scheduling periods for each store.

Main responsibilities:

- Start date
- End date

- Active period state
- Closed period state

Staff availability submissions are linked to a period.

availabilities

Stores original staff availability submissions.

Main responsibilities:

- Store ID
- Period ID
- Staff name
- Shift date
- Start time
- End time

Important design rule:

Original availability records are treated as submitted source data and should not be overwritten by Auto Shift V1.

Store Operation Settings

store_business_hours

Stores default business hours for each store.

Main responsibilities:

- Default open time
- Default close time

store_business_overrides

Stores date-specific business-hour overrides.

Main responsibilities:

- Store-specific special business days
- Closed days
- Date-specific open / close times

japanese_holidays

Stores Japanese public holidays used for required staffing logic.

store_required_staff_settings

Stores required staffing rules for each store.

Main responsibilities:

- Weekday required staff count
- Weekend required staff count
- Holiday required staff count
- Peak time range
- Day-specific required count settings
- Shortage detection thresholds

This table is used for shortage detection and Auto Shift V1 input.

Auto Shift V1 Tables

store_auto_shift_settings

Stores store-level Auto Shift settings.

Main responsibilities:

- Opening coverage window
- Closing coverage window

These settings define how long before or after business hours the system checks opening and closing coverage.

auto_shift_staff_settings

Stores Auto Shift-specific staff capabilities.

Main responsibilities:

- Whether a staff member can handle opening
- Whether a staff member can handle closing

This allows Auto Shift V1 to distinguish normal availability from opening / closing responsibility.

auto_shift_runs

Represents one Auto Shift generation run.

Main responsibilities:

- Store ID
- Period ID
- Algorithm version
- Run status
- Input snapshot
- Score summary

Each Auto Shift generation creates one run record.

auto_shift_drafts

Stores the generated draft result for an Auto Shift run.

Main responsibilities:

- Run ID
- Store ID
- Period ID
- Generated draft JSON

The generated draft is stored separately from the manager-confirmed result.

confirmed_shift_snapshots

Stores the final manager-confirmed schedule snapshot.

Main responsibilities:

- Run ID
- Store ID
- Period ID
- Confirmed schedule JSON
- Diff summary
- Manager reason tags
- Confirmation timestamp

This table allows the system to compare the generated draft with the final manager-confirmed schedule later.

Logging Tables

shift_change_logs

Stores shift edit logs.

Main responsibilities:

- Store ID
- Period ID
- Staff name
- Shift date
- Old start / end time
- New start / end time
- Changed role
- Changed timestamp

export_logs

Stores Excel export logs.

Main responsibilities:

- Store ID
- Period ID
- Export type

- Shift count
- Staff count
- Export timestamp

Design Principles

1. Store-based data separation

Most operational tables include `store_id`.

This makes it possible to keep each manager limited to their own store data.

2. Preserve original submissions

Submitted availability data is treated as original source data.

Auto Shift V1 does not overwrite original staff submissions.

3. Separate generated drafts and confirmed schedules

Auto Shift V1 separates:

- Original availability data
- Generated draft data
- Manager-confirmed final schedule
- Confirmed snapshot

This reduces operational risk and makes later analysis possible.

4. Support future improvement

By storing both generated drafts and confirmed schedules, the system can later analyze how managers edit Auto Shift results.

This data can be used to improve Auto Shift V2.